DAY 1 · LECTURE 1 · 45 MINUTES

# The Wreckage

*Linux Basics & the Birth of Containers*

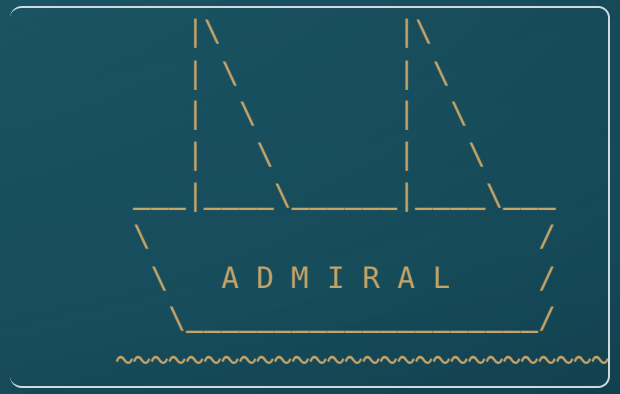
*The mainframe is gone. Time to learn what we're standing on.*

# Where we are

- This morning you built a **ship**: one Ubuntu VM, one `setup-client.sh`, an identical environment for every sailor in the crew.
- The next 45 minutes: what that ship actually **is** — and the lighter craft we build on top of it.
- Four short legs:
  - Why the old ship sank
  - The anatomy of a Linux machine
  - How a container actually works
  - How we build one
- Then we hit the water: **Lab 01 — The First Raft**.

PART I

# Why the SS Legacy Sank




> *"...then all collapsed, and the great shroud of the sea rolled on as it rolled five thousand years ago." — Herman Melville, Moby-Dick*

# The SS Legacy

> *"His ship, the SS Legacy, was built around a mainframe which made it slow to change course."*

- One enormous ship. One mainframe. Every function bolted to the same hull.
- This is a **monolith** — all the code and all its dependencies, shipped and run as a single unit.
- It floated for years... right up until it had to change.

# What a monolith costs you

- **Change is risky** — edit one feature, rebuild and retest *everything*.
- **Failure spreads** — one leak floods the whole hull.
- **Scaling is blunt** — need more of one part? Launch another entire ship.
- **"Works on my machine"** — the environment is implicit and never written down.

*Faculty translation: thirty laptops, thirty slightly different ships — and an IT queue you don't control. One bootstrap script routes around both.*

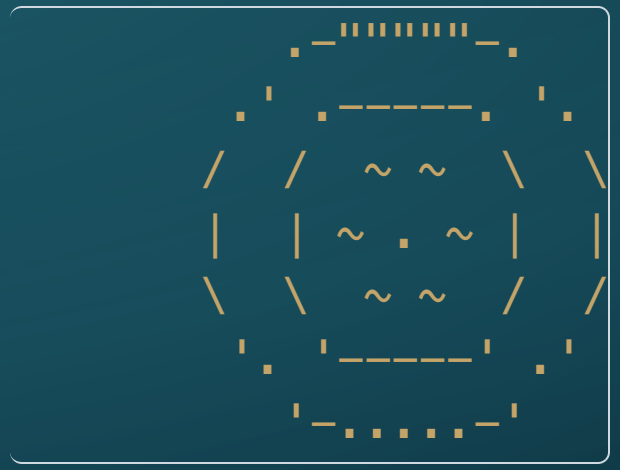
# A new approach

> *"They broke the work down into smaller problems, creating tiny, easily replaceable tools and building supplies."*

- Stop trying to build one unsinkable ship.
- Build **many small craft** — each does one job, each replaceable without dragging the rest down.
- Those small craft have a name: **containers**.
- The rest of today is learning to build and sail them.

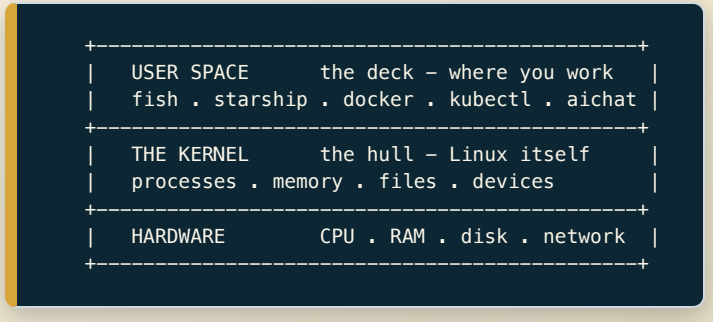
PART II

# Anatomy of Your Vessel



> *"Know every plank of your hull, or the sea will teach ye — the hard way." — the Boatswain*

# Every Linux system has two halves




- The **kernel** runs the hardware. **User space** is every program you launch.
- Hold this two-layer picture — the whole container idea rides on it.

# The Kernel — the hull

- The privileged core of the operating system. You stand on it; you never touch it directly.
- Its job:
  - talk to the **hardware**
  - **schedule** which program runs when
  - hand out **memory**
  - own the **filesystem**
  - enforce who is allowed to do what
- Exactly **one** kernel runs the machine. Everything on board shares it.

*Remember that last point — it is the entire reason containers are light.*

# User Space — the deck

- Everything above the kernel — every program the crew actually runs.
- Everything `setup-client.sh` installed this morning lives here: Fish, Starship, Docker, `kubectl`, `aichat`.
- Programs can't poke the hardware directly. They **ask the kernel**, through **system calls**.
- An app, plus the libraries and files it needs to run = a **user-space environment**.

# Everything is a process

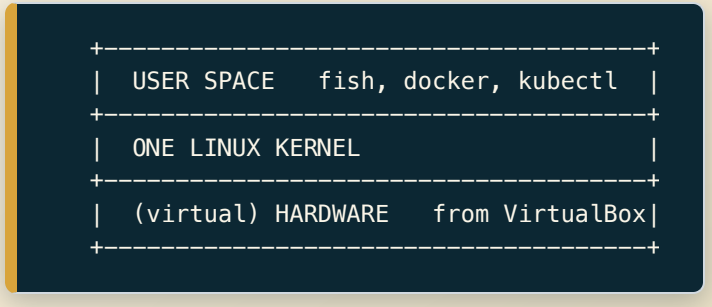
- A **process** is simply a running program.
- `setup-client.sh` was a process. The shell you are typing into is one right now.
- Each process has an **ID**, an **owner**, and its own slice of **memory**.
- The kernel decides who runs and when — see it yourself with `ps aux` or `top`.

> **Hold this thought:** *a container is just a process. We will come straight back to it.*

# Everything is a file

- In Linux, **almost everything is a file** — documents, directories, even devices and disks.
- It all hangs off a **single tree**, starting at the root: `/`.
- Every file carries **permissions** — who may read, write, or execute it.
- A container gets its **own private view** of this tree. That is the trick we are building toward.

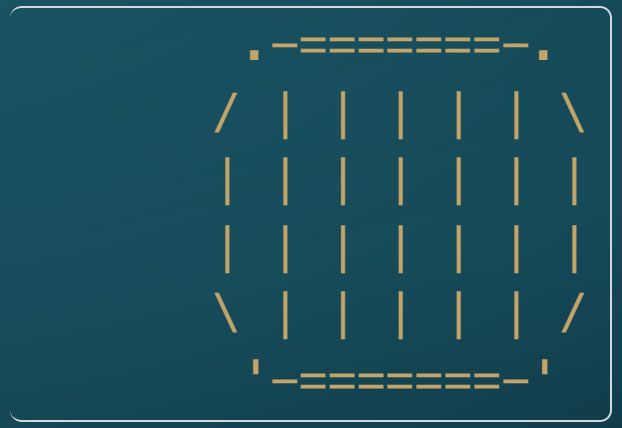
# So — what did you build this morning?




- Virtual hardware → **one** Linux kernel → a user space full of tools.
- Keep this on screen. Containers do **not** replace any of it.
- They **reuse the kernel** and add a fresh, isolated user space on top.

PART III

# From Hull to Crates

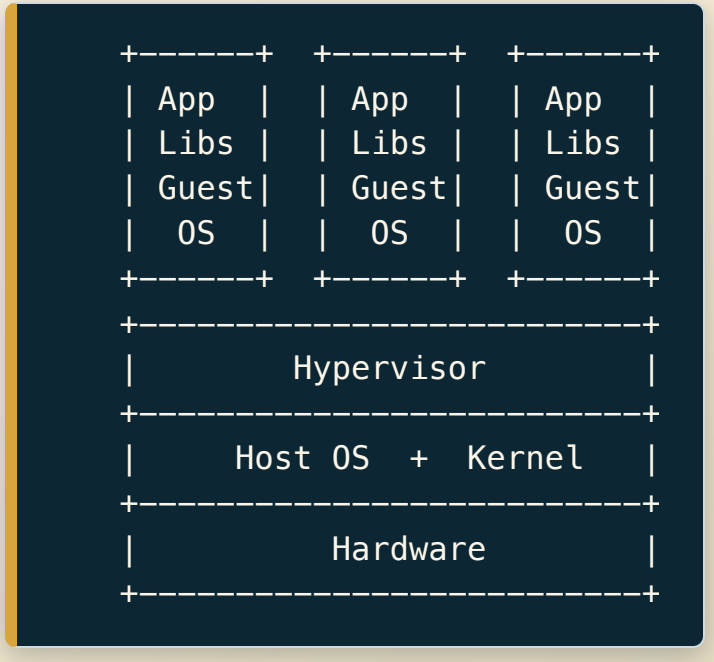


> *"Why haul a whole ship when a crate'll do? Pack light, sail fast." — the Boatswain*

# The real problem

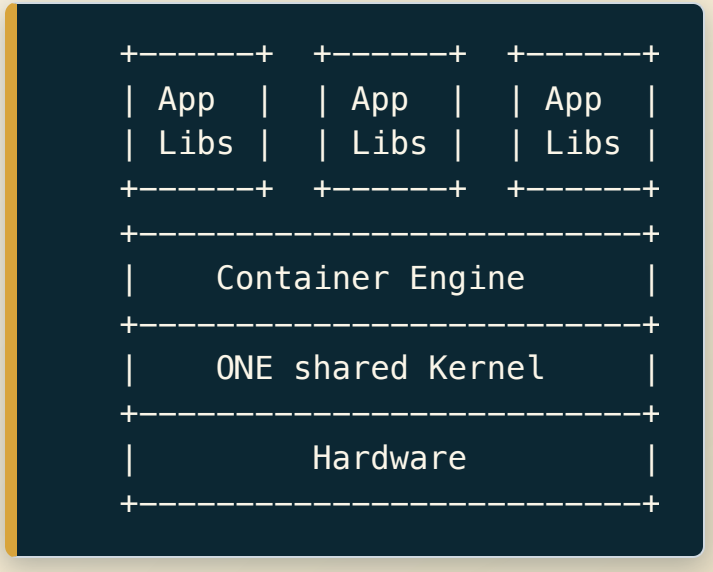
- Shipping your app's **code** is easy. Shipping the **environment it needs** is the hard part.
- App A needs library version 1. App B needs version 2. Same ship — now they fight.
- "*Works on my machine*" really means: my environment differs from yours, and **neither is written down**.
- We need to package the app **together with its whole user space**.

# First instinct: give every app its own ship




- It works — full isolation. But every ship hauls a **whole Guest OS**:
- gigabytes of disk · minutes to boot · **its own kernel to patch**.

# The insight: share the hull




- Does each app really need its own **kernel**? **No.**
- It needs its own **user space** — its own files and libraries.
- A **container** is an isolated user space running on the **host's** kernel.

# Isolation: Linux namespaces

- A container is just a process — so how is it walled off? With **namespaces**.
- Namespaces control what a process is allowed to **see**:
  - its own **process list**
  - its own **filesystem** tree
  - its own **network**
  - its own **hostname**
- Inside the crate, the program is convinced it is alone on the ship.

# Limits: Linux cgroups

- Namespaces decide what a process can **see**. **Control groups** (cgroups) decide what it can **use**.
- A ceiling on **CPU time**, **memory**, and **I/O** — enforced by the kernel.
- One greedy crate can't hog the whole deck and capsize its neighbours.
- Namespaces + cgroups are both **ordinary kernel features**. Nothing exotic — Linux already had them.

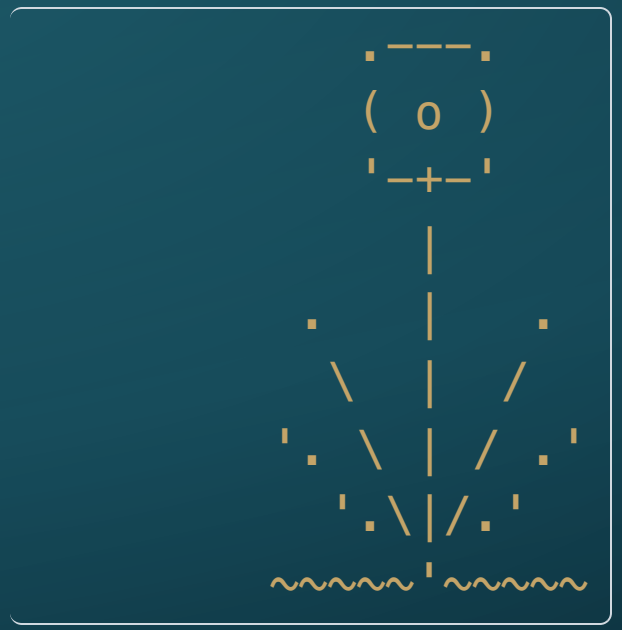
# So what *is* a container?

- A container is **not a tiny virtual machine**.
- It is: a normal **Linux process** + **namespaces** (what it sees) + **cgroups** (what it uses) + its own packaged **user space** — all riding the **shared host kernel**.
- No Guest OS. No boot sequence. It starts in **milliseconds**.

> **The Boatswain:** "*A container's just a wooden crate, lad. The hull is the OS, the loading docks are the ports — the crate only carries the cargo.*"

PART IV

# Building a Crate



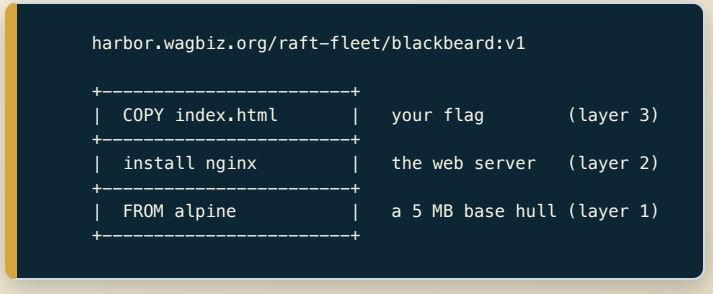

> *"Measure twice, cut once — a crate built right is a crate built once." — the Boatswain*

# Image vs. Container

- **Image** — the blueprint *and* the parts: a frozen, **read-only** package of a user space.
- **Container** — a **running instance** of an image.
- One image → **many** identical containers. Build once, run anywhere, the same every time.

*The blueprint stays in the drawer. The crates go in the water.*

# Images are built in layers




- Each instruction in the recipe adds **one layer**, stacked bottom-up.
- Layers are **cached and shared** — change only your `index.html`, and only the top layer rebuilds.

# The Dockerfile — the blueprint

- The blueprint is a plain-text file: the `Dockerfile`.
- A short recipe — for example, `FROM` a base image, then `COPY` your files in.
- Written once; it produces the **exact same image** on any machine, anywhere.

> *We won't drill the syntax on a slide. In* **Lab 01**, *the* **Socratic Boatswain** *— your AI crewmate — coaches you through writing it yourself.*

# The Harbor — where images live


- A finished image needs a home: a **registry** stores and shares images.
- Your island's registry is **Harbor**, at `harbor.wagbiz.org`.
- Build it, push it to the docks, and the cluster can pull it whenever it needs to.

# Instructor Superpower

## THE REPRODUCIBLE ENVIRONMENT

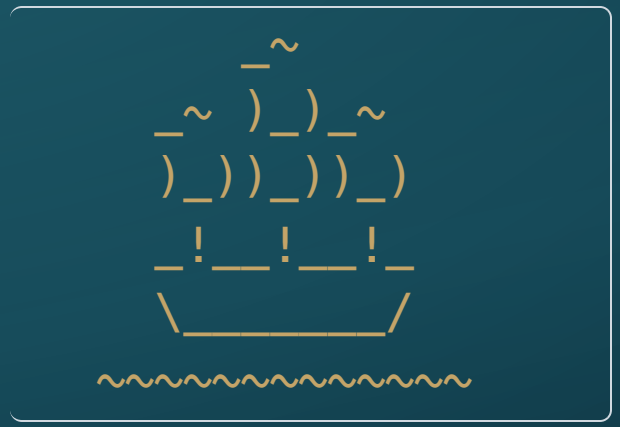
- This morning's `setup-client.sh` and this afternoon's container are the **same idea at two scales**:
- *describe the environment as code → build it once → hand everyone an identical copy.*
- Containerize an assignment and you ship students the **environment**, not just the code — thirty identical setups from one definition.
- The environments you can choose to support have radically expanded. You don't wait for IT to bless a toolchain — you build it into the image yourself.

# Up next: Lab 01 — The First Raft

- Write a `Dockerfile` for an Nginx web server.
- Hoist your flag — a custom `index.html` with your name on it.
- **Build** the image, **test** that your raft floats, **push** it to Harbor.
- Meet your AI crewmate: the **Socratic Boatswain** (it will *not* hand you the code).

> *Success looks like one thing: your raft afloat and docked in the Harbor — ready for the cluster to find it this afternoon.*

# To the rafts.



*The hull is yours. Now build something that floats.*